

Who Should Be in the Loop?

Allocating Expert Review for Qualitative Coding with Language Models

Anonymous ACL submission

Abstract

Large language models can assist qualitative coding, but specialist review does not scale easily. We present **WarrantRoute**, a source-grounded protocol for testing which reviewer expertise is needed for particular qualitative judgments and for allocating review under a fixed expert-time budget. The planned study will compare researchers without specialist expertise, qualitative methods experts, and domain experts on natural outputs and controlled variants targeting unsupported evidence, source concentration, counterevidence loss, contextual flattening, and unsupported abstraction. It will evaluate whether a policy frozen before held-out testing can route items without hiding less frequent or negative cases, and whether selected feedback repairs defects without introducing new ones. The current manuscript reports the protocol and a measurement-pipeline qualification on entirely fictional packets; it reports no real-corpus model, rater, routing, or repair result.

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1 Introduction

Large language models (LLMs) can suggest codes, organize codebooks, and apply qualitative categories across large collections of text (Dai et al., 2023; Wang et al., 2026; Matveyenko et al., 2026). Researchers often review these outputs, edit suggestions, explain disagreements, and guide later analysis. Most evaluations still treat their judgments as one general source of human feedback, even when the task calls for different kinds of knowledge.

A qualitative methods expert may recognize that a topic label has been presented as an interpretive theme. A domain expert may notice that an analysis has lost a cultural distinction or omitted a clinically important exception. A researcher without either specialization may still identify an unsupported

quotation or an overly broad claim. A single average score conceals these differences and provides little guidance about how to use limited expert time.

Evaluations of LLM supported clinical thematic analysis remain fragmented across expert review and automatic similarity measures (Yi et al., 2025a). The usual measures do not resolve this problem. More than one code space may be defensible (Chen et al., 2026), agreement between humans and LLMs may not predict blind expert preference (Liu et al., 2026), and model confidence can be poorly calibrated (Li et al., 2026). Audits have also found missing participant evidence, summaries presented as quotations, and failures that depend on local context (Parkington et al., 2025; Bang et al., 2026; Subbiah et al., 2026). Provenance identifies the source of a passage. It does not show that the passage supports the interpretation or represents the range of views in the corpus.

The planned experiment treats review as an *allocation problem*: who should examine a qualitative interpretation and its supporting evidence? **WarrantRoute** will collect judgments from researchers without specialist expertise, qualitative methods experts, and domain experts. The study will include natural model outputs and controlled variants with known defects. A simple policy will send each item to no specialist, a qualitative methods expert, a domain expert, or both. The policy will use information available before specialist review, and all comparisons will receive the same amount of expert time. We will freeze the policy before testing it on a new domain and model family. Figure 1 summarizes the framework.

The planned study is designed to contribute:

1. a source-grounded benchmark that isolates five validity problems without assuming a single correct thematic interpretation;
2. a human-evaluation design that adapts evidential credibility and plural-reference cover-

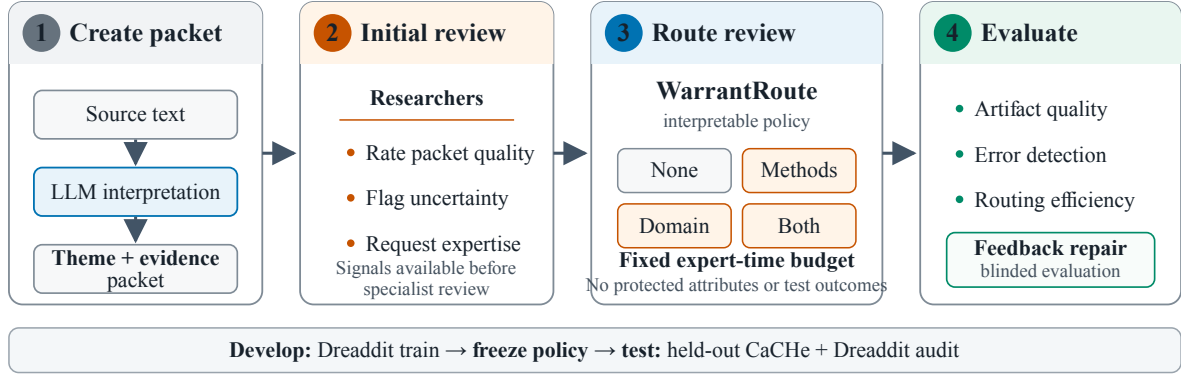


Figure 1: Overview of the planned WarrantRoute evaluation. An interpretable policy will route source-grounded theme and evidence packets to researchers alone, a qualitative methods expert, a domain expert, or both within a fixed time budget. The routing comparison uses CaChe and Dreddit; KODIS, CaSiNo, and AMI remain supplementary, unassigned corpus lanes.

age from prior work and adds study-specific, source-grounded operationalizations of voice-and-boundary preservation and scope calibration; and

3. a planned held-out domain test that compares several allocation rules at equal cost and measures whether selected feedback repairs a defect without creating a new one.

The present manuscript reports the protocol and a synthetic measurement-pipeline qualification, not a real-corpus human-evaluation result.

2 Related Work

Human feedback in qualitative analysis.

Human-in-the-loop systems let analysts revise codes and rationales (Dai et al., 2023), pair-code and discuss disagreements (Gao et al., 2024), convert repeated feedback into persistent instructions (Wang et al., 2026), and retain positionality and revision histories (Ye et al., 2026). Multi-agent systems extend this workflow: Auto-TA automates source-to-theme generation (Yi et al., 2025c), SFT-TA uses supervised fine-tuned agents (Yi et al., 2025b), and TAMA incorporates cardiac expert guidance (Xu et al., 2026). These studies establish scalable collaboration, but their evaluations emphasize alignment, usability, control, or aggregate output quality rather than reviewer specialization.

Review allocation and evaluator dependence.

Selective-review methods route items using confidence and ensemble diversity (Zhao and Liu, 2025)

or rationale disagreement (Tajik et al., 2026), although confidence can be miscalibrated or anchor-sensitive (Li et al., 2026). Evaluator identity also changes conclusions: blind expert verification can diverge from agreement metrics (Liu et al., 2026), and specialists who read source material can differ from nonspecialists who see isolated themes (Moskalewicz et al., 2026). Adjacent biomedical work evaluates generated protocol pseudocode against a generated reference (Yi et al., 2024), but not the source warrant of qualitative interpretations. Prior work therefore studies allocation signals and evaluator perspective separately, not which expert role a source-linked packet requires under matched expert time.

Source grounding and artifact quality.

Provenance systems preserve links among quotations, codes, themes, and revisions (Ye et al., 2026; Yi et al., 2026), yet models can omit human-identified excerpts, substitute summaries for quotations, or produce paraphrased, truncated, and hallucinated evidence (Parkington et al., 2025; Bang et al., 2026). Computational thematic-analysis evaluations have operationalized credibility through human acceptance and semantic alignment and, later, binary LLM judgments (Qiao et al., 2025; Wang et al., 2026); OpenCodes instead measures coverage relative to a pooled code space (Chen et al., 2026). These constructs are complementary: cited evidence may support a claim that omits dissent, while broad coverage does not ensure appropriate scope. WarrantRoute therefore adapts evidential credibility and plural-reference coverage, adds packet-level operationalizations of voice-and-

boundary preservation and scope calibration, and reports all four separately. The open comparison is whether methods and domain expertise detect complementary source-grounded failures, and whether expertise-aware routing improves detection over expertise-agnostic allocation at the same cost.

3 Task and Study Design

3.1 Evidence packets and controlled variants

The planned unit of analysis is a **theme and evidence packet**. Each future real-data packet will contain a qualitative claim, a short explanation, and exact excerpts only from an access-controlled, pseudonymized analysis copy after the required two-person privacy and permissions review, with project-pseudonymous speaker or document identifiers, local context, and a summary of source coverage. Raters will judge whether the evidence supports the claim and preserves important differences, counterevidence, and source voice. The task is a bounded source-warrant and representation assessment. It will not assess a complete reflexive thematic analysis or assume that one thematic account is universally correct (Braun and Clarke, 2021a,b).

The study will sample natural LLM outputs and construct variants vetted by experts. Each variant will change one property frozen in the preregistration:

1. **unsupported evidence**: a topically related excerpt does not warrant the claim;
2. **source concentration**: wording implies breadth while evidence comes disproportionately from one participant or document;
3. **counterevidence loss**: a rare, negative, or boundary case is removed;
4. **contextual flattening**: a consequential cultural or domain specific qualification is erased; or
5. **unsupported abstraction**: a local observation is generalized beyond its evidence.

An independent construction panel will check the original packet, the intended defect, and the absence of a second manipulation. Pilot items will not enter the main analysis, and each rater will see at most one version of a base packet.

3.2 Artifact-quality measures

We will preregister four artifact-quality measures, separately from the routing and repair outcomes below. Two adapt measures used in prior computational thematic analysis. **Evidential credibility** asks whether the core interpretive meaning is directly supported by the cited evidence in local context. Unlike prior binary LLM judgments (Qiao et al., 2025; Wang et al., 2026), the planned primary measure is a blinded human rating. **Plural-reference coverage** is the percentage of concepts in a frozen reference space pooled from multiple independent analysts that are recovered by at least one generated code or theme (Chen et al., 2026). The pool will preserve alternative concepts rather than designate one analyst’s themes as truth, and blinded experts will adjudicate many-to-many conceptual matches.

We propose two study-specific, packet-level operationalizations. **Voice-and-boundary preservation** assesses whether consequential differences, dissent, counterevidence, less-frequent or boundary cases, and contextual qualifications survive synthesis. **Scope calibration** assesses whether claim breadth, strength, polarity, and causal language are proportionate to the supplied evidence. Raters will score these two measures and evidential credibility from 1 to 5. The analysis will retain the ordinal scores and report ratings of 4 or 5 as descriptively adequate; “cannot judge” will be reported separately. Plural-reference coverage will be calculated only on complete natural-output inventories, separately for codes and themes, and will not be treated as a 1–5 rating. We will interpret the four measures only after deterministic quotation, attribution, assignment, and provenance checks. Because coverage is recall-like, we will report generated-concept volume and human-audited unsupported or redundant concepts so that verbosity cannot masquerade as quality. We will not combine the four measures into a composite. Full anchors and matching rules appear in Appendix B.4.

3.3 Datasets and leakage controls

The project maintains five candidate corpus lanes that differ in domain, interaction structure, and provenance. Dreddit’s official training split is provisionally assigned to development, its official test split to a secondary in-domain audit, and the 11 CaChe focus groups to the planned held-out cross-domain confirmatory evaluation. KODIS is

a received, aggregate-audited private exploratory dialogue lane. CaSiNo and the AMI scenario-meeting subset remain source-inventory candidates. KODIS, CaSiNo, and AMI have no frozen experimental role, approved model or rater pathway, or sampling eligibility. Here, “held out” describes the planned procedure, not model-training provenance.

Table 1 reports each lane at its current preparation stage. Dreddit’s candidate pool applies duplicate and exact-date cluster quarantines; Dreddit and CaChE remain pending manual privacy review and project authorization. KODIS counts exclude 775 human–AI dyads and nine human–human rows without transcripts, and its exploratory parser retains repeated-role turns. CaSiNo counts exclude 2,378 structured negotiation events, while AMI counts transcribed speech segments and words. All lanes are English-language or translated into English. The heterogeneous units are not common experimental denominators, and unassigned rows are not sampling eligible. Definitions, provenance, access conditions, and calculations appear in Appendix B.1.

Under the planned protocol, only Dreddit training data will inform prompts, packet rules, controlled variants, routing features, and thresholds. We will freeze these choices before test generation, preserve complete source groups, quarantine exact training duplicates that overlap Dreddit test text, and cluster CaChE analyses by focus group and transcript-local speaker. KODIS, CaSiNo, and AMI will not inform these choices while their experimental roles remain unassigned. Packets will use the same context limits across adopted evaluation datasets. We will retain prompts, decoding settings, run identifiers, source spans, and revisions. Protected text will be processed only under approved terms. The complete leakage and generation protocol appears in Appendix B.1.

3.4 Raters and annotation

The planned study will recruit three groups. **Qualitative methods experts** will have demonstrated experience conducting qualitative analysis. **Domain experts** will have research, professional, or relevant lived experience in the source domain. **Researchers without specialist expertise** will be able to apply the study rubric but will not meet either expert definition for the assigned item. Dual expertise will be recorded as a separate status. Item construction, initial rating, policy development, and final adjudication will be handled by different people

wherever staffing permits.

All raters will complete the same tutorial and review balanced subsets in a randomized incomplete-block design. For each packet they will rate evidential credibility, voice-and-boundary preservation, scope calibration, and confidence on scales from 1 to 5. The common quality anchors will be 1 (not adequate), 2 (major problems), 3 (partially adequate; material revision needed), 4 (mostly adequate; minor issues only), and 5 (fully adequate); confidence will use separate uncertainty anchors. They will also choose accept, revise, reject, or escalate; indicate whether they cannot judge the item; provide a short rationale and, if needed, request a particular kind of expertise. The system will record review time. “Cannot judge” will remain a separate response category, with no conversion to a midpoint or a quality failure. Qualitative methods experts and domain experts will submit independent ratings before adjudication. When their disagreement reflects defensible interpretations, the item will remain plural or ambiguous.

We will choose the final sample size after a separate pilot by simulating the crossed structure of raters and packets. The preregistration will state the number of unique raters, base packets, ratings per group, exclusions, and the smallest effect or noninferiority margin that the design can support.

4 Routing by Expertise

The planned routing method will use information available before specialist review: researchers’ ratings of evidential credibility, voice-and-boundary preservation, and scope calibration, plus disagreement, confidence, abstention, the requested kind of expertise, response time, and nonsensitive provenance measures such as excerpt count, source concentration, and quotation exactness. Plural-reference coverage will be evaluation-only because it relies on the frozen analyst reference space. The router will not use manipulation labels, expert scores from the test set, protected participant attributes, or test outcomes.

We will fit an interpretable regularized model that chooses among no escalation, review by a qualitative methods expert, review by a domain expert, or review by both. We will select its thresholds on the development data to maximize recall of serious errors within a fixed budget measured in expert minutes. We will then compare five conditions:

1. **researchers only**: no specialist review;

Dataset	Context	Language	Inventory	Stage
Dreaddit (Turcan and McKeown, 2019)	Social-media stress posts	English	2,929 posts; 3,553 segments; 3,516 candidate; 301,279 words 11 groups; 3,944 turns;	Processed; review pending
CaCHe (Mehta, 2024)	Sexual and reproductive health focus groups	Translated English	3,076 eligible; 98,735 words	Processed; review pending
KODIS (Hale et al., 2025)	Customer-service dispute dialogues	English	2,085 human dyads; 2,076 dialogues; 27,121 messages; 568,333 words 1,030 dialogues; 11,919 messages;	Exploratory; review pending
CaSiNo (Chawla et al., 2021)	Campsite negotiation dialogues	English	228,675 words 138 meetings; 69,258 segments;	Source only
AMI scenario (Carletta et al., 2006)	Product-design meetings	Spoken English	793,764 words	Source only

Table 1: Dataset statistics.

2. **both expert groups:** a qualitative methods expert and a domain expert review every item;
3. **random:** route the same budget at random;
4. **uncertainty only:** route using confidence, disagreement, and abstention; and
5. **WarrantRoute:** predict both whether escalation is needed and which role is relevant.

Every condition will use the same observed expert time. The planned confirmatory outcomes on the CaCHe cross-domain evaluation are recall of serious errors, false acceptance, and expert minutes per correctly escalated error; the same outcomes will be reported on the Dreaddit in-domain audit as secondary results. These policy-level outcomes are distinct from the four artifact-quality measures. A secondary experiment will use one fixed revision model and compare no feedback, feedback from researchers, feedback selected by WarrantRoute, and feedback from both expert groups. A separate blinded panel will judge whether the revision fixes the target defect and whether it creates a new unsupported claim, omission, or distortion.

5 Evaluation and Analysis

The planned analysis addresses three questions. First, for which errors do qualitative methods experts or domain experts differ materially from researchers without specialist expertise? Second, can the initial ratings identify those cases within a fixed budget of expert time? Third, does the selected feedback repair defects in a new domain and model family?

For the first question, separate cumulative-link mixed-effects models will analyze evidential credibility, voice-and-boundary preservation, and scope calibration; logistic mixed-effects models will analyze decisions and defect detection. Models will include packet version, evaluator group, error family,

domain, and their preregistered interactions, with random effects for rater and base packet. Plural-reference coverage will be estimated separately at the complete-run level, not from sampled packets or controlled twins. For the second question, we will compare policies on the cross-domain evaluation with cluster-bootstrap intervals and a preregistered noninferiority contrast against review by both expert groups. For the third question, a logistic mixed model will relate successful repair to feedback condition and error family. Dreaddit results will be secondary; the CaCHe analysis will become confirmatory only after its manifest and analysis plan are frozen and preregistered.

We will report effect sizes and uncertainty intervals. Ordinal Krippendorff’s α , Gwet’s AC2, raw agreement, and confidence calibration will describe measurement behavior; they will not establish validity on their own (McDonald et al., 2019). We will freeze and preregister confirmatory contrasts, multiplicity, and the CaCHe packet manifest before generating model outputs for that corpus. Development and audit results, comparisons of natural and controlled items, and other robustness checks will appear in Appendix D. Table 8 is reserved for reliability, and Table 9 for transfer and routing-feature ablations. Confidence and router calibration will be reported separately in the appendix.

6 Results

This section separates the completed synthetic measurement-pipeline qualification from the planned real-corpus reporting shell. The qualification used four fictional templates from an earlier corpus design, including KODIS-shaped and CANDOR-shaped proxies. It tests whether the four measures can be calculated reproducibly; it provides no evidence about Dreaddit, CaCHe, KODIS, CaSiNo, AMI, or any other real corpus. Replace every later blank from a single frozen analysis ex-

port and reconcile all denominators before writing the final abstract. Report the CaChe confirmatory results before the Dreaddit audit and development performance.

6.1 Synthetic measurement-pipeline qualification

We applied the paper-aligned measures to 60 outputs generated from entirely fictional packets: five recent model variants, four corpus-shaped packets, and three independent runs per model and packet. Three blinded model judges independently scored evidential credibility, voice-and-boundary preservation, and scope calibration using the manuscript anchors. For this engineering pilot, the benchmark’s frozen designed constructs instantiate the reference space; each judge matched every reference separately to generated codes and themes, and a two-of-three panel rule determined recovery. In Table 2, the ordinal columns report the percentage of 36 blinded ratings per model scored 4 or 5, and code and theme coverage report panel-majority recovery across 63 frozen reference opportunities per level and model. Brackets are 95% packet-by-run cluster-bootstrap intervals, and integrity is deterministic. All judges and candidate models are from one provider, so these are measurement-pipeline results rather than real-corpus or human-evaluation results.

The pilot exposes a ceiling effect that would be hidden by reporting adequacy alone. GPT-5.6 Sol, GPT-5.5, and GPT-5.6 Terra received 100% adequacy on all three ordinal measures, while their deterministic integrity pass rates were 10/12, 9/12, and 5/12. Code coverage ranged only from 93.7% to 95.2%, and theme coverage from 85.7% to 93.7%. We therefore retain the four constructs as the source-grounded artifact profile but do not use them alone for model selection: integrity failures are exclusionary, full ordinal distributions and controlled-defect sensitivity are reported, and coverage is accompanied by output-volume and unsupported-concept diagnostics. Under that gate-first rule, GPT-5.6 Sol remains the remediation candidate rather than a frozen paper model.

6.2 Pre-human corpus checkpoint

All five real corpora are now present and deterministically inventoried. Dreaddit and CaChe have validated restricted derivatives; KODIS has a private exploratory turn conversion; and CaSiNo and AMI have verified source inventories but no adopted

adapters. These preparation statistics populate Table 1, not performance outcomes. The remaining result cells require independent ratings, analyst reference construction and concept matching, observed review time, or blinded repair judgments.

6.3 Study accounting

The final manuscript will report [N TOTAL RATERS] recruited raters and [N RATINGS] analyzed judgments over [N PACKETS] packets. [REPORT EXCLUSIONS, BALANCE, AND THE TEST SET]. No corpus-size statistic can fill these fields; Table 3 begins at the human-feedback boundary and is reserved for study counts by experimental role.

6.4 Artifact-quality profile

The final manuscript will report CaChe credibility, voice-and-boundary, and scope-calibration adequacy rates, plus code and theme plural-reference coverage, only after the confirmatory analysis is frozen and run. It will report these measures with the integrity prerequisite, denominators, and uncertainty, and provide full ordinal distributions, cannot-judge rates, matching uncertainty, unmatched reference concepts, and generated-concept volume in Appendix B.4.

6.5 Where expertise changes judgments

[SUMMARIZE THE INTERACTION BETWEEN EVALUATOR GROUP AND FAILURE FAMILY, INCLUDING UNCERTAINTY]. Table 4 reports predicted probabilities of detecting a defect. [STATE WHETHER THE TWO EXPERT GROUPS PROVIDED COMPLEMENTARY, REDUNDANT, OR INCONCLUSIVE JUDGMENTS].

6.6 Routing under matched expert time

After the expert-minute budget is preregistered and the study is run, the final manuscript will report [PRIMARY TEST RESULT], [EFFECT VERSUS UNCERTAINTY ROUTING WITH 95% CONFIDENCE INTERVAL], and [NONINFERIORITY RESULT VERSUS BOTH EXPERT GROUPS]. Table 5 is reserved for the five routing policies under the matched expert-time budget. Results must be reported by failure family if an average gain conceals misses on negative cases.

Model	Integrity pass	Evidential adequacy	Voice/boundary adequacy	Scope adequacy	Code coverage	Theme coverage	Serious error
GPT-5.6 Sol	10/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	93.7 [89.1, 98.4]	85.7 [78.5, 93.4]	0.0 [0.0, 0.0]
GPT-5.4	9/12	100.0 [100.0, 100.0]	97.2 [91.7, 100.0]	100.0 [100.0, 100.0]	93.7 [89.1, 98.4]	93.7 [89.1, 98.4]	5.6 [0.0, 13.9]
GPT-5.5	9/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	93.7 [89.1, 98.4]	92.1 [87.3, 96.8]	0.0 [0.0, 0.0]
GPT-5.6 Luna	8/12	80.6 [69.4, 91.7]	80.6 [69.4, 91.7]	88.9 [80.6, 97.2]	95.2 [90.9, 100.0]	85.7 [79.7, 91.9]	8.3 [0.0, 16.7]
GPT-5.6 Terra	5/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	95.2 [90.9, 100.0]	88.9 [84.1, 93.7]	0.0 [0.0, 0.0]

Table 2: Synthetic measurement-pipeline qualification.

Quantity	Dev.	In domain	Cross domain
Source participants/documents	—	—	—
Base packets / controlled twins	—	—	—
Researchers without specialist expertise	—	—	—
Qualitative methods experts	—	—	—
Domain experts	—	—	—
Independent ratings	—	—	—
Median review minutes	—	—	—

Table 3: Planned reporting shell for study counts by experimental role. Raters and packets will be reported as unique counts.

6.7 Does routed feedback repair defects?

The final manuscript will report [RATE AND CONFIDENCE INTERVAL] for packets repaired by WarrantRoute feedback without a new error, alongside [RESEARCHER FEEDBACK RATE] for researcher feedback and [BOTH EXPERT GROUPS RATE] for feedback from both expert groups. Table 6 is reserved for repair outcomes, new errors, credibility changes, and expert time. [REPORT PERFORMANCE RELATIVE TO EXPERT MINUTES AND THE MOST IMPORTANT FAILURE CASE].

7 Discussion

[STATE WHICH EVALUATOR GROUP ADDED VALUE FOR WHICH FAILURE TYPES, WITH EFFECTS]. [STATE WHETHER THE TWO EXPERT GROUPS PROVIDED COMPLEMENTARY JUDGMENTS]. [REPORT THE PRACTICAL COST AND EXPERT-REVIEW COVERAGE CONSEQUENCE AT THE OBSERVED REVIEW TIMES]. Any resulting conclusions must be limited to the evidence-review task, failure families, and evaluator groups actually studied.

On the new domain and model family, [STATE THE TRANSFER RESULT AND ITS INTERPRETATION]. [STATE WHAT THE RESULT IMPLIES FOR ROUTING BEYOND GENERIC UNCERTAINTY]. A useful review queue should preserve the reason for escalation and allow a reviewer to request another form of exper-

tise. Persistent disagreement remains visible for deliberation.

8 Limitations

Any resulting estimates will reflect the recruited reviewers, whose judgments may reproduce disciplinary, cultural, or institutional assumptions. Controlled variants will isolate individual failure mechanisms, but naturally occurring errors may be subtler or combine several defects. Repeated judgments from the same people may increase precision without increasing the diversity of expertise. Routing can also create false reassurance around items that are not escalated, so the design includes a random audit and failure-family-specific false-acceptance reporting. The task covers bounded evidence packets. A complete qualitative analysis also requires familiarization, memoing, reflexivity, and accountable interpretation. Generalization will be limited to English Reddit stress segments and English translations of focus groups from western Kenya, as well as [MODEL FAMILIES AND RECRUITMENT SAMPLE]. If the unassigned dialogue lanes are later adopted, KODIS and CaSiNo cannot support cross-dialogue participant counts or participant-disjoint splits from their public releases, while AMI scenario meetings are elicited role play and transcript-only processing omits prosody, timing, overlap, and multimodal evidence.

9 Ethics Statement

Real text model processing and rater exposure begin only after a documented institutional determination and all source, model provider, rater access, retention, and release gates are satisfied; source collection approvals do not transfer. We treat all corpora as potentially identifying. Analysis copies are minimized and restricted, protected text goes only to approved services, and every excerpt shown to raters or released must pass a documented two-person privacy and permissions review. Raters participate voluntarily outside supervisory or evaluative relationships, with identifiers stored separately.

Failure family	Researchers	Methods experts	Domain experts	Largest contrast
Unsupported evidence	_____	_____	_____	_____
Source concentration	_____	_____	_____	_____
Counterevidence loss	_____	_____	_____	_____
Contextual flattening	_____	_____	_____	_____
Unsupported abstraction	_____	_____	_____	_____

Table 4: Predicted probability of detecting each failure, with 95% confidence intervals, by evaluator group.

Policy	Expert minutes	Serious error recall	False acceptance	Difference from both experts	Errors per expert hour
Researchers only	0	_____	_____	_____	Not applicable
Random	_____	_____	_____	_____	_____
Uncertainty routing	_____	_____	_____	_____	_____
WarrantRoute	_____	_____	_____	_____	_____
Both expert groups	_____	_____	_____	Reference	_____

Table 5: Planned reporting shell for routing performance on the CaChe confirmatory evaluation at matched budgets measured in expert minutes. Intervals will use a bootstrap clustered by base packet.

Feedback	Repair	New error	Credibility change	Expert minutes
None	_____	_____	_____	0
Researchers	_____	_____	_____	0
WarrantRoute	_____	_____	_____	_____
Both expert groups	_____	_____	_____	_____

Table 6: Planned reporting shell for blinded revision. A repair will count as successful only if it fixes the targeted defect without introducing an unsupported claim.

Public artifacts contain only aggregate, synthetic, or cleared material. Routing supports research review, not clinical or participant decisions. Details appear in Appendix C.

10 Conclusion

WarrantRoute is designed to measure how reviewer expertise affects the detection and repair of unsupported qualitative interpretations under a fixed review budget. The current manuscript establishes the protocol and reports only the fictional-packet measurement qualification; [STATE THE PRIMARY REAL-CORPUS RESULT AND ITS PRACTICAL IMPLICATION ONLY AFTER THE FROZEN STUDY IS COMPLETE].

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	The appendix is intentionally skeletal until the pro-	821
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	page main paper.	824

B Full Study Protocol

B.1 Corpus provenance, preprocessing, and leakage controls

The project documents five candidate corpus lanes. Dreddit’s official train partition is provisionally assigned to development, its official test partition to a secondary in-domain audit, and all 11 CaChe focus groups to the planned held-out cross-domain confirmatory evaluation. KODIS is prepared only for a private, unsplit exploratory role. CaSiNo and the AMI scenario-meeting subset remain source-inventory dialogue candidates. KODIS, CaSiNo, and AMI have no frozen manuscript experiment role, approved model or rater pathway, completed contextual privacy review, or sampling eligibility.

Dreddit is a publicly downloadable English Reddit corpus for stress detection. Its released labeled subset contains 3,553 human annotated segments from 2,929 posts across ten subreddits in five domains (Turcan and McKeown, 2019). The archive provides official train and test partitions and post identifiers, but no author identifier. It therefore supports segment to post provenance and document concentration checks, not participant counts or participant level claims. The author archive contains 2,838 train segments from 2,343 posts and 715 test segments from 586 posts. It contains no README or dataset license. We retain text, the stress label, subreddit, and sentence range in the restricted analysis copy. We omit original post identifiers, timestamps, engagement measures, the internally inconsistent confidence field, and unused lexical features. Project specific identifiers preserve post grouping without exposing the released post identifier. The official splits have no post identifier overlap, but three exact texts occur in both partitions under different identifiers and conflicting labels. We retain those rows for audit but make the three training copies and 18 later within train exact duplicates ineligible for sampling, leaving 2,817 development segments and 715 audit segments before privacy quarantine. A checked exact-date rule then quarantines 13 complete development post clusters and three audit clusters, leaving 2,804 development and 712 audit candidate segments, or 3,516 segments and 301,279 whitespace delimited words in total. The remaining text still requires contextual privacy review.

The CaChe deposit contains 11 participant-numbered and selectively redacted focus-group transcripts about how COVID-19 affected school-

ing, sexual behaviour, and sexual and reproductive health in western Kenya (Mehta, 2024). The source study reports six groups with 54 adolescent girls and young women and five with 53 community men. Sessions used English, Swahili, Dholuo, or a mixture and were translated into English (Awiti et al., 2024). The deposit is version 1, DOI 10.25417/uic.26495884.v1, and CC BY 4.0. A fixed checksum allowlist verifies all 11 source files. Processing retains 3,575 participant turns and 369 collective turns; 3,076 participant turns meet the earlier three-word eligibility rule, but short and collective turns are no longer deleted and remain available as context. Each record has a stable focus-group and turn identifier, a transcript-local speaker identifier, source line spans, source hashes, translation status, the preceding moderator question, and prior-turn references. Normalization yields 106 observed transcript-local participant keys, one fewer than the 107 participants reported by the source paper; we do not claim complete participant coverage until this is reconciled. Conflicting source dates are recorded but excluded from model and router features.

The received KODIS workbook contains 2,860 unique dyads and 107 columns (Hale et al., 2025). Its 2,085 human-human rows include 2,076 nonempty dialogues with 27,121 parsed messages and 568,333 words; the remaining nine have no transcript. Across all human-human rows, 1,712 are marked Resolution and 373 Impass; among retained transcripts the counts are 1,708 and 368. The median nonempty dialogue has 12 messages (interquartile range 11–14) and 242 words (interquartile range 180–334). The private exploratory builder excludes all 775 human-AI rows and all workbook surveys, demographics, country, preferences, tactics, point totals, raw dyad identifiers, and timestamps. It preserves nine seller-first dialogues and 271 dialogues with at least one consecutive same-role transition rather than imposing artificial alternation. All retained records have the single role exploratory_unsplit; the release has no stable cross-dialogue participant identifier, so it cannot support participant-disjoint splitting or participant-level denominators. This exploratory conversion is not a frozen manuscript experiment or a substitute for contextual privacy review.

The acquired CaSiNo source is the official CC BY 4.0 repository snapshot pinned to commit 2f6ed4a. It contains 1,030 dyadic campsite negotiation dialogues, 11,919 ordinary messages, 2,378

structured negotiation events, and 228,675 whitespace delimited words in the ordinary messages (Chawla et al., 2021). The source paper reports 846 participants, but the public JSON provides only two dialogue local role tokens and no stable cross-dialogue participant token. We therefore reject the supplied 900/30/100 random split for WarrantRoute, do not reconstruct linkage from participant profiles or text, and leave the corpus unassigned unless trustworthy provider linkage becomes available or the entire corpus is preregistered in one unsplit role after the governance gates are satisfied. Any future adapter will retain ordinary dialogue text and turn lineage only. It will exclude structured events, annotations, demographics, personality measures, preferences, free text reasons, negotiation outcomes, and raw identifiers.

The acquired AMI manual annotation artifact contains a selectively extracted source inventory for 138 elicited scenario meetings in 35 meeting series. The four person meetings are English product design discussions. The selected files contain 793,764 transcribed words, 69,258 speech segment elements, and mappings for 140 global source speaker identifiers. These form 35 meeting and speaker connected components, 33 with four meetings and two with three meetings (Carletta et al., 2006). No audio, video, signals, slides, questionnaires, demographics, participant file, or nontranscript annotation category was extracted. The official filename labels the manual annotations as version 1.6.2, while the embedded README says release 1.7; we bind the source by its exact URL, filename, size, and checksum rather than asserting that those labels are equivalent. The corpus is not sampling eligible. A future transcript adapter must discard timing and raw source identifiers, and all 35 meeting and speaker components must remain indivisible in any adopted split.

For Table 1, a source is one Reddit post, focus group, KODIS human-human dyad, CaSiNo dialogue, or AMI meeting. Dreddit source units are released labeled segments, and its candidate units exclude duplicate texts and complete exact-date post clusters. CaChe retained units include participant and collective turns, while its eligible count applies the earlier three word participant-turn rule. KODIS units are parsed role messages from nonempty human-human transcripts. CaSiNo units are ordinary source messages and exclude structured events. AMI units are upstream speech segment elements. KODIS, CaSiNo, and AMI counts

are not sampling eligible or common analysis denominators.

Under the planned protocol, Dreddit’s official train partition will be the only corpus used to develop prompts, packet rules, controlled variants, routing features, and thresholds. After these choices are frozen, its official test partition will provide an in-domain audit. All 11 CaChe focus groups will be held out from WarrantRoute development for the planned cross-domain confirmatory evaluation. “Held out” refers to the experimental procedure, not model-training provenance. The deposit is public and published themes are known, so we will not call it unopened or model-unseen or use the published themes as scoring gold.

The power analysis and expert time budget determine a common number of base packets for the Dreddit audit and CaChe evaluation. Sampling preserves post clusters in Dreddit and focus group and transcript local speaker clusters in CaChe. Both corpora use identical packet and context limits. Dreddit labels and published CaChe themes may stratify audits but are not treated as thematic ground truth or router features. KODIS, CaSiNo, and AMI enter no manuscript generation, sampling, or model selection procedure unless a later preregistration assigns them an experimental role after corpus specific validation and governance approval.

If the governance gates are completed, we will generate packets using [MODEL VERSIONS] and [GENERATION PROCEDURE]. We will retain prompts, decoding settings, context order, run identifiers, source spans, and human revisions and will not select test outputs after inspecting their quality. We will use text or transcripts only, process them under the documented approved governance arrangement, and not send protected material to an external model unless the access terms and our institution permit it. Material released publicly will be synthetic or specifically cleared for release.

[CURRENT INSTITUTIONAL DETERMINATION; SOURCE, PLATFORM, MODEL PROCESSING, AND RATER ACCESS AUTHORIZATIONS FOR DREADDIT, CaChe, KODIS, CASINO, AND AMI; KODIS AND CASINO UNSPLIT ROLE DECISIONS; AMI COMPONENT SPLIT; RATER EXCERPT RULES; FINAL PACKET COUNTS; MODEL SNAPSHOTS; PROMPTS; DECODING SETTINGS; RUN COUNTS; AND FROZEN MANIFEST HASHES].

B.2 Role definitions and recruitment

[ELIGIBILITY FOR RESEARCHERS WITHOUT SPECIALIST EXPERTISE, QUALITATIVE METHODS EXPERTS, AND DOMAIN EXPERTS; HANDLING OF DUAL EXPERTISE; RECRUITMENT; COMPENSATION; POSITIONALITY; EXCLUSIONS; AND INDEPENDENCE SAFEGUARDS].

B.3 Complete annotation instrument

The three packet-level quality questions are: (1) *How fully does the cited evidence support the core interpretive meaning in context?* (evidential credibility); (2) *How well does the packet preserve consequential differences, dissent, counterevidence, less-frequent or boundary cases, and contextual qualifications?* (voice-and-boundary preservation); and (3) *How well do the strength and breadth of the claim, including causal or group-level language, match the supplied evidence?* (scope calibration).

For each question, 1 means not adequate, 2 major problems, 3 partially adequate and requiring material revision, 4 mostly adequate with only minor issues, and 5 fully adequate. The tutorial will supply construct-specific examples at each anchor. For rating r_{ijm} from rater i on packet j , we will analyze the full ordinal response; the descriptive adequacy rate for measure m will be the proportion of evaluable rater–packet judgments scored 4 or 5:

Evidential credibility. A 1 indicates absent, unverifiable, misattributed, or contradictory evidence; a 3 indicates that the core meaning is partly supported but a material link or qualifier lacks evidence; and a 5 indicates that every material part of the core interpretation is directly warranted by sufficient contextual evidence.

Voice-and-boundary preservation. A 1 indicates that a consequential voice, counterexample, or contextual distinction is erased or reversed; a 3 retains the main pattern but underplays a meaningful viewpoint or boundary; and a 5 retains all consequential variation, dissent, less-frequent cases, and contextual qualifications visible in the packet.

Scope calibration. A 1 indicates that claim breadth, polarity, or causal strength is incompatible with the evidence; a 3 has the broadly appropriate direction but requires a material change in scope or strength; and a 5 fully

matches participant-, group-, and corpus-level scope and makes no stronger causal or prevalence claim than the evidence permits.

Scores 2 and 4 represent the corresponding intermediate judgments under the common anchors. Raters select “cannot judge” when the supplied excerpts, local context, or source-coverage summary are insufficient for the specified dimension.

$$A_m = \frac{\sum_{i,j} \mathbb{1}(r_{ijm} \in \{4, 5\})}{\sum_{i,j} \mathbb{1}(r_{ijm} \text{ is evaluable})}. \quad (1)$$

“Cannot judge” is neither a midpoint nor a failure and is reported with its own denominator. Confidence has separate anchors from 1 (very uncertain) to 5 (very certain). Exact-string and source-offset checks are recorded as deterministic diagnostics; raters judge whether verified excerpts warrant the interpretation in context.

[CONSENT TEXT, TUTORIAL EXAMPLES, COMPREHENSION CHECKS, ACCEPT/REVISE/REJECT/ESCALATE WORDING, ABSTENTION RULE, RATIONALE PROMPT, AND TIMING DISCLOSURE].

B.4 Artifact-quality definitions and matching

Plural-reference coverage will be calculated only for complete, natural-output code and theme inventories, not from sampled packets or controlled twins. Before inspecting test outputs, multiple analysts will independently produce code and theme concepts. We will freeze their pooled reference space after merging only concepts adjudicated to be equivalent, retaining the originating analyst and source provenance. This pool will represent a bounded set of defensible concepts available to the study team, not universal thematic ground truth.

For output level $\ell \in \{\text{code}, \text{theme}\}$, let P_ℓ be the frozen pooled concepts and G_ℓ the generated concepts. Independent experts blinded to system condition may make many-to-many conceptual matches, denoted by $p \sim g$. Coverage is

$$C_\ell = \frac{|\{p \in P_\ell : \exists g \in G_\ell, p \sim g\}|}{|P_\ell|}. \quad (2)$$

We will report code and theme coverage separately with preregistered uncertainty intervals, list unmatched reference concepts, retain disagreements about matching, and not count unmatched generated concepts as errors without separate evidential review. To diagnose coverage gained through indiscriminate output, we will also report generated

code and theme counts, unmatched generated concepts, split and merge patterns, redundancy, and the human-audited unsupported-concept rate. Because the measure relies on the analyst reference space, it will not be available to the router. The three ordinal measures and plural-reference coverage will remain separate; no omnibus quality score will be computed.

For controlled variants, the primary manipulation check will map unsupported evidence to evidential credibility; source concentration, counterevidence loss, and contextual flattening to voice-and-boundary preservation; and unsupported abstraction to scope calibration. We will also report cross-construct effects rather than assume that each defect can affect only one measure. Plural-reference coverage will be restricted to natural-output inventories and will not serve as a manipulation check for controlled twins.

B.5 Packet construction and assignment

[NATURAL ITEM SAMPLING, CONSTRUCTION OF CONTROLLED VARIANTS, VETTING, MANIPULATION CHECKS, RANDOMIZATION, INCOMPLETE BLOCK ASSIGNMENT, AND BLINDING].

B.6 Power and confirmatory analysis

[PILOT VARIANCE COMPONENTS, SIMULATION DESIGN, POWER TARGET, NON-INFERIORITY MARGIN, MODEL EQUATIONS, MULTIPLICITY PLAN, FALLBACKS, AND EXCLUSIONS].

B.7 Routing implementation

[FEATURE DEFINITIONS, PREPROCESSING, MODEL, HYPERPARAMETERS, BUDGET SELECTION, THRESHOLDS, CROSS FITTING, FREEZE DATE, AND TEST ACCESS LOG].

C Detailed Ethics and Data Governance

C.1 Source review and access conditions

Approval of an original collection does not authorize this secondary study, model processing, or disclosure to raters. Before real-text model processing or rater disclosure, we will record our institution’s determination, its scope and date, and the governing source, platform, and access terms for each corpus.

The received KODIS README records collection dates, field definitions, and a required scholarly citation, but it does not state a data license, secondary-use consent, or authorization for external model processing or rater disclosure. We therefore keep the exploratory output local and do not treat receipt of the workbook as permission for a manuscript experiment (Hale et al., 2025).

The CaChe source study reports written consent and review by Rush University Medical Center (22011311), Maseno University Ethics Review Committee (MSU/DRPI/MUERC/01021/21), and University of Illinois Chicago (2022-0220), with a favorable opinion from Liverpool School of Tropical Medicine (21-087) (Awiti et al., 2024). The CaSiNo paper reports institutional review, informed consent covering collection and later use, a right to withdraw, and removal of MTurk and HIT identifiers before public release (Chawla et al., 2021). Although that paper calls the released data anonymized, we do not adopt that label: dialogue and rich participant metadata can remain identifying. The AMI site documents European Commission ethics review and signed consent that anticipated possible release to the wider research community; its consent form states that anonymity could not be guaranteed and allowed participants to request removal (AMI Corpus, n.d.). The CC BY 4.0 licenses for CaSiNo and AMI do not grant privacy or personality rights or authorize this project’s model processing, rater disclosure, retention, or release. We did not find an ethics determination, Reddit author consent, privacy procedure, or dataset license in the Dreddit paper or author archive (Turcan and McKeown, 2019); any use therefore remains conditional on a documented institutional and platform basis.

C.2 Project safeguards and rater protections

We do not describe the corpora as anonymous. Dreddit retains searchable verbatim posts. CaChe contains sensitive sexual and reproductive health narratives that may remain identifying despite participant numbers and selective redaction. KODIS combines role-play dialogue with country, survey, preference, tactics, and outcome fields; the private exploratory output retains only dialogue text, minimal row lineage, role, and aggregate diagnostics. CaSiNo raw records combine dialogue with demographics, personality measures, preferences, free text reasons, and outcome surveys; those metadata remain excluded from any future working output.

AMI source XML contains transcript text, precise timing, and attribute bearing source identifiers; future working outputs must omit timings and raw identifiers. Restricted analysis copies use project pseudonymous identifiers, minimize fields, mask direct identifier patterns, and preserve protected linkage information separately. Pattern masking does not replace contextual privacy review. Protected text is not sent to a model or service unless its processing, retention, and training terms are approved. [STATE STORAGE LOCATION, ACCESS ROLES, RETENTION PERIOD, DELETION PROCEDURE, APPROVED MODEL PROVIDERS, AND PROVIDER RETENTION OR TRAINING SETTINGS].

Every excerpt shown to raters or considered for release will receive a two-person privacy and permissions review. Public artifacts will contain aggregate results, code, prompts, schemas, synthetic packets, and only text or ratings specifically cleared for release. We will not publish raw hashes of short sensitive excerpts without assessing linkage risk. Raters will receive a study-specific consent form and privacy warning, may withdraw under the approved procedure, and will participate voluntarily outside supervisory, employment, grading, or authorship decisions. Identifiers will be stored separately from responses. [STATE RECRUITMENT CHANNELS, ELIGIBILITY, COMPENSATION AND RATE JUSTIFICATION, WITHDRAWAL HANDLING, RATER DEMOGRAPHICS, AND FINAL CONSENT LANGUAGE]. Routing is a research review aid, not a clinical or participant-level decision system; random audits and false-acceptance reporting address, but do not remove, the risk of false reassurance.

D Supplementary Results

The source-grounded artifact profile below is now part of the compiled results shell. Its cells remain blank because corpus inventories cannot substitute for human ordinal ratings, reference-concept matching, or serious-error judgments. KODIS, CaSiNo, and AMI receive rows only if a later pre-registration assigns an experimental role.

[ADD CONFUSION MATRICES, SELECTIVE-RISK VERSUS EXPERT-REVIEW-COVERAGE CURVES, RESULTS BY FAILURE FAMILY, COMPARISONS OF NATURAL AND CONTROLLED ITEMS,

CONFIDENCE AND ROUTER CALIBRATION PLOTS, RATIONALE ANALYSIS, AND ROBUSTNESS CHECKS].

E Reproducibility and Governance Checklist

- ☐ Confirm institutional, source or platform, model processing, and rater access conditions for Dreddit, CaChe, KODIS, CaSiNo, and AMI; confirm unsplit roles for KODIS and CaSiNo if adopted; validate AMI transcript preprocessing and freeze component isolated splits; then freeze the analytic contract, prompts, model snapshots, and manifests.
- ☐ Obtain ethics determination and finalize voluntary recruitment and withdrawal procedures that avoid coercion.
- ☐ Freeze role criteria, item manifest, the four metric definitions and anchors, plural-reference construction and matching rules, assignment, exclusions, and power analysis.
- ☐ Freeze router features, thresholds, budgets measured in expert minutes, primary outcomes, and analysis code before generating CaChe model outputs.
- ☐ Release only packets cleared by the two-person excerpt review, permitted ratings, prompts, code, model and run metadata, non-text manifest hashes, and text-derived hashes only when source terms permit and linkage-risk review clears them, plus a datasheet.
- ☐ Replace every \fillin, \blankcell, and \blankwide; set \predatadraftfalse; verify the eight page boundary; and update the abstract last.

Corpus / evaluation role	System	Complete inventories	Rated packets	Integrity pass n/N	Evidential adequacy	Voice/boundary adequacy	Scope adequacy	Code coverage	Theme coverage	Cannot judge	Serious error
CaCHe / cross-domain confirmatory	Frozen primary	—	—	—	—	—	—	—	—	—	—
CaCHe / cross-domain confirmatory	Frozen comparator	—	—	—	—	—	—	—	—	—	—
Dreaddit / in-domain audit	Frozen primary	—	—	—	—	—	—	—	—	—	—
Dreaddit / development	Selected primary	—	—	—	—	—	—	—	—	—	—

Table 7: Source-grounded artifact profile. Adequacy is the percentage of evaluable packet ratings scored 4 or 5; “cannot judge” has a separate denominator. Plural-reference coverage is computed separately for complete code and theme inventories. Integrity requires schema and assignment completeness, quotation/source fidelity, valid provenance edges, and cross-source support for cross-source themes. Report 95% intervals clustered by the preregistered sampling unit. Synthetic qualification results are excluded.

Evaluator group and construct	Krippendorff α	Gwet AC2	Raw agreement	Cannot judge	Median time
Researchers: credibility	—	—	—	—	—
Researchers: voice/boundary	—	—	—	—	—
Researchers: scope calibration	—	—	—	—	—
Methods experts: credibility	—	—	—	—	—
Methods experts: voice/boundary	—	—	—	—	—
Methods experts: scope calibration	—	—	—	—	—
Domain experts: credibility	—	—	—	—	—
Domain experts: voice/boundary	—	—	—	—	—
Domain experts: scope calibration	—	—	—	—	—

Table 8: Reliability shell for the three ordinal artifact-quality measures. Report intervals and score prevalence in the final appendix; confidence and router calibration are reported separately.

Routing configuration	Dreaddit dev.	Dreaddit audit	CaCHe	Recall change	Expert minutes
Full features	—	—	—	—	—
No disagreement	—	—	—	—	—
No abstention	—	—	—	—	—
No provenance	—	—	—	—	—
Confidence only	—	—	—	—	—

Table 9: Effects of transfer and removal of routing features. Define the reported metric in the final caption.